FINAL REPORT

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Project Title	DESIGN OF A REUSABLE HIGH-ALTITUDE RADIATION MEASUREMENT SYSTEM

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INTRODUCTION

As humanity's presence extends further into space, understanding radiation at high altitudes has become increasingly critical. This project aims to explore radiation dynamics using high-altitude balloons positioned approximately 35 kilometers above Earth's surface. With the continual expansion of satellite projects, space exploration endeavors, and missions to celestial bodies like the Moon and Mars, comprehending the radiation environment in space and the upper atmosphere is paramount.

High-altitude balloons offer a practical and cost-effective method for investigating cosmic radiation. Central to this research are two fundamental questions: how does radiation vary across different altitudes from Earth's surface up to 35 kilometers, and how can we elucidate the Regener-Pfotzer maximum—an atmospheric event that marks the peak intensity of secondary cosmic radiation?

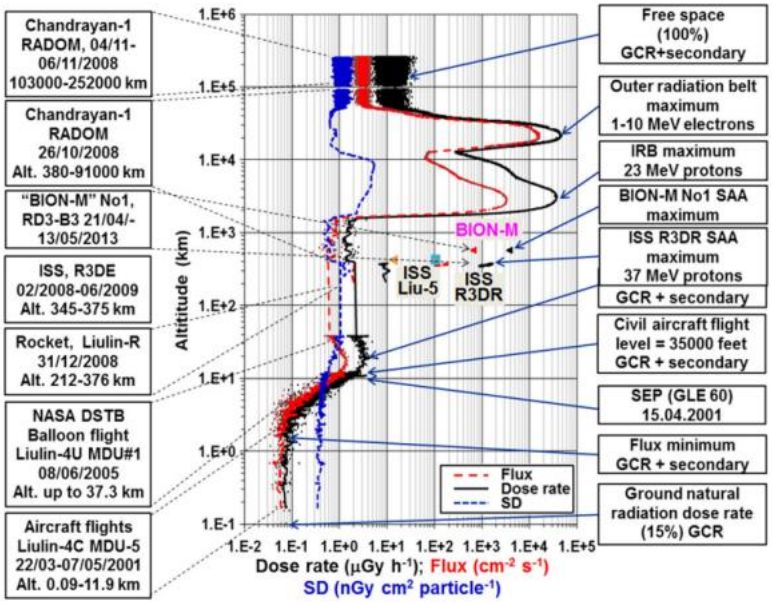


Figure 1:Variations of the absorbed dose rate, flux, and specific dose for altitude range from 0.1 to 250.000km [5]

The Regener-Pfotzer maximum, named after Erich Regener and Georg Pfotzer, represents the altitude at which the intensity of cosmic radiation is at its highest due to the interaction of primary cosmic rays with the Earth's atmosphere, resulting in a peak in secondary radiation. This peak typically occurs at altitudes between 15 to 25 kilometers. Studies have shown that the altitude of the Regener-Pfotzer maximum can vary based on several factors, including solar activity and geomagnetic conditions [1] [16]. For instance, during periods of high solar activity, the altitude of the maximum can be slightly lower due to increased solar radiation interacting with the atmosphere [16].

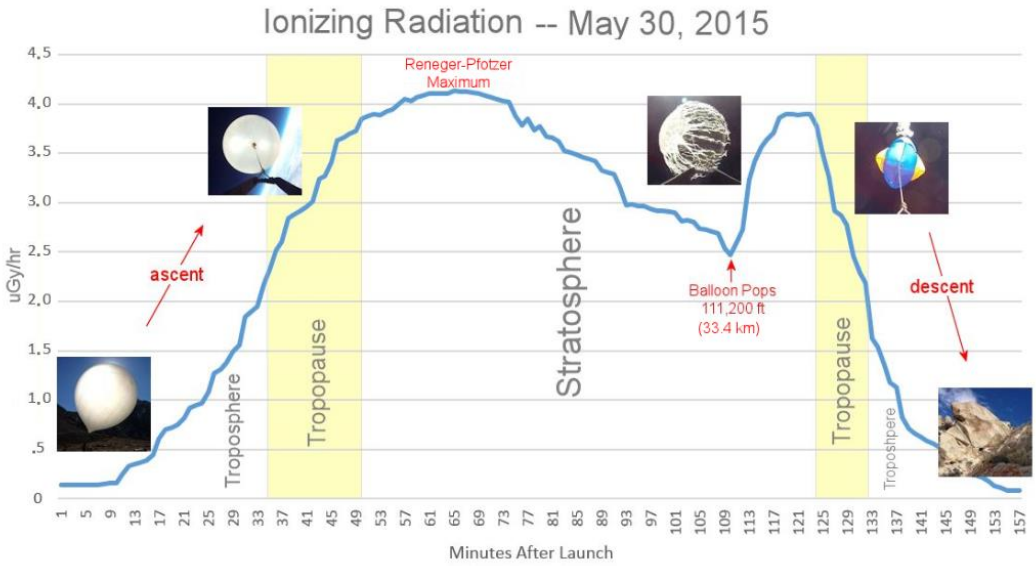


Figure 2:Dose Rate vs Time graph and The Regener-Pfotzer maximum. [22]

The exploration of high-altitude radiation boasts a rich historical lineage with various platforms such as satellites, balloons, aircraft, and sounding rockets contributing significantly to our knowledge base. Early experiments in cosmic ray research, such as those conducted by Victor Hess in the early 20th century, utilized balloons to establish the presence of ionizing radiation from outer space, laying the groundwork for modern radiation studies [5]. Recent studies have investigated the latitudinal effects on the Regener-Pfotzer maximum, providing insights into how these radiation peaks vary with geomagnetic conditions [1].

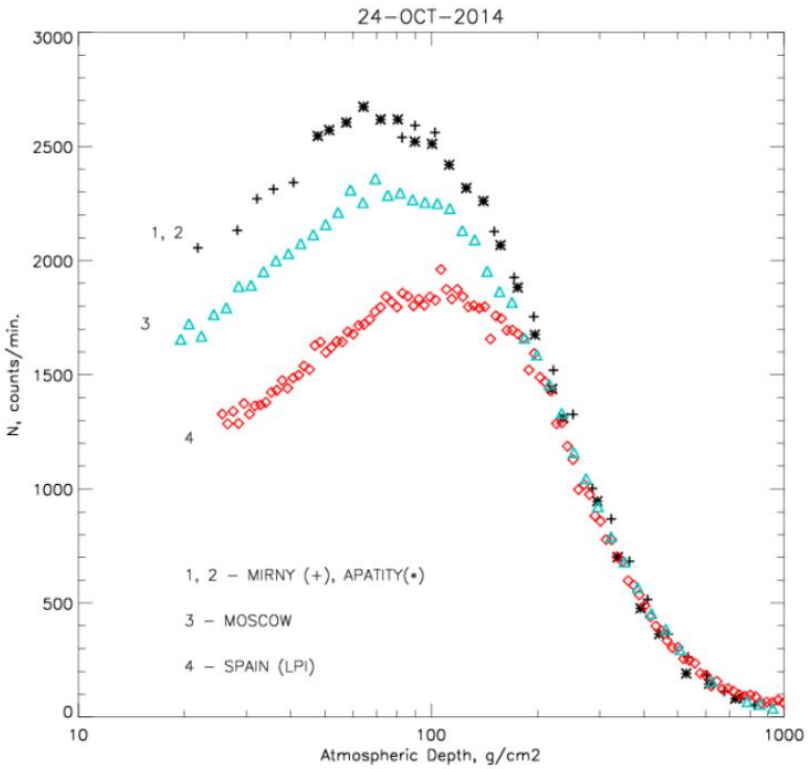


Figure 3:Cosmic Ray Measurements at Different Geomagnetic Vertical Cut-Off Rigidities [15]

During the total solar eclipse on August 21, 2017, researchers investigated potential changes in the Regener-Pfotzer maximum's altitude due to the eclipse. Measurements taken before and during the eclipse indicated that the Regener-Pfotzer maximum occurred at approximately 20 kilometers, with no significant changes in altitude observed during the eclipse [16]. This study highlights the stability of the Regener-Pfotzer maximum despite varying atmospheric conditions. Additionally, long-duration high-altitude measurements have confirmed the consistency of the Regener-Pfotzer maximum. These measurements are crucial for understanding the impact of cosmic radiation on the Earth's atmosphere and for developing radiation protection strategies for both human spaceflight and high-altitude aviation [5] [16].

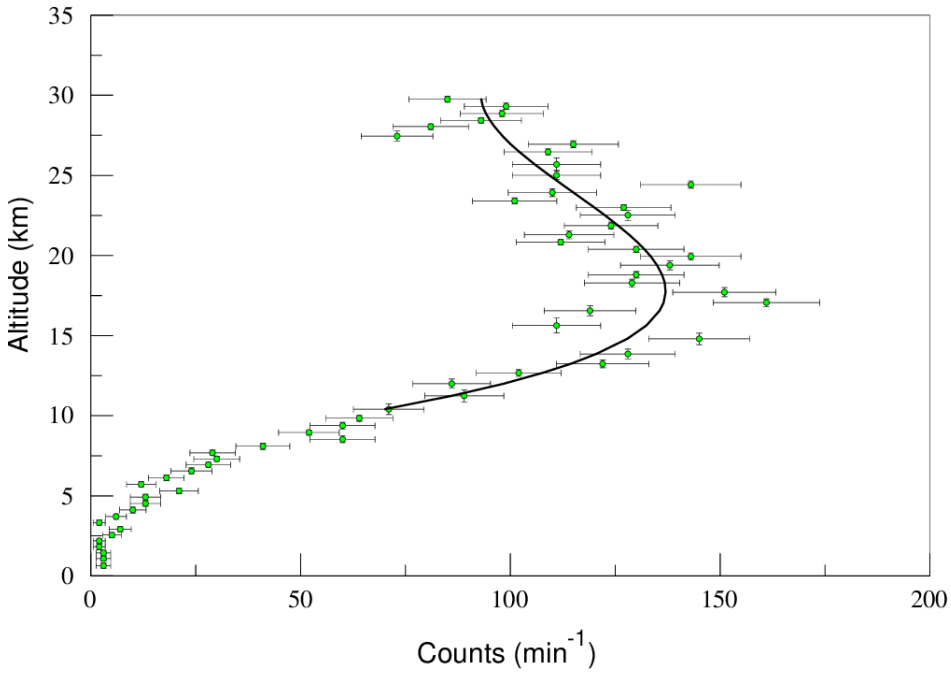


Figure 4: Altitude versus omnidirectional counts of particles during Solar cycle. [16]

Recent technological advancements have significantly enhanced the capabilities of high-altitude radiation measurement systems. [9] developed a compact radiation monitor incorporating a Geiger-Müller counter and silicon detectors designed for radiation measurements on Turkish space rockets. This monitor demonstrated its capability to detect the Regener-Pfotzer maximum at approximately 20 kilometers altitude. Moreover, the integration of advanced communication technologies such as LoRa for long-range, low-power data transmission has improved the reliability and efficiency of data collection from high-altitude platforms [8] [24]. The implementation of Linear Kalman Filters has further enhanced the precision of height estimation, crucial for accurate radiation profiling [14].

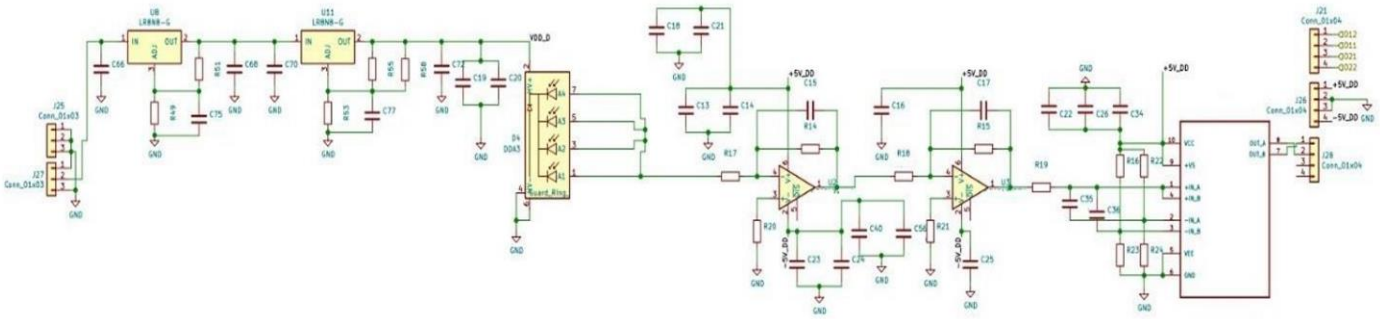


Figure 5: Schematic of the compact radiation monitor developed by Demirköz et al. [9]

For this project, we aim to measure dose rates across various altitudes, anticipating an increase in radiation intensity as the altitude increases, peaking at the Regener-Pfotzer maximum. At sea level,

the dose rate is expected to be relatively low, around 0.1 μSv/h. As the balloon ascends, this rate is expected to increase, reaching approximately 10 μSv/h at altitudes around 15-25 kilometers, where the Regener-Pfotzer maximum occurs. Beyond this altitude, the dose rate is expected to decrease again. Understanding these variations is crucial for assessing the radiation environment faced by high-altitude aviation and future space missions.

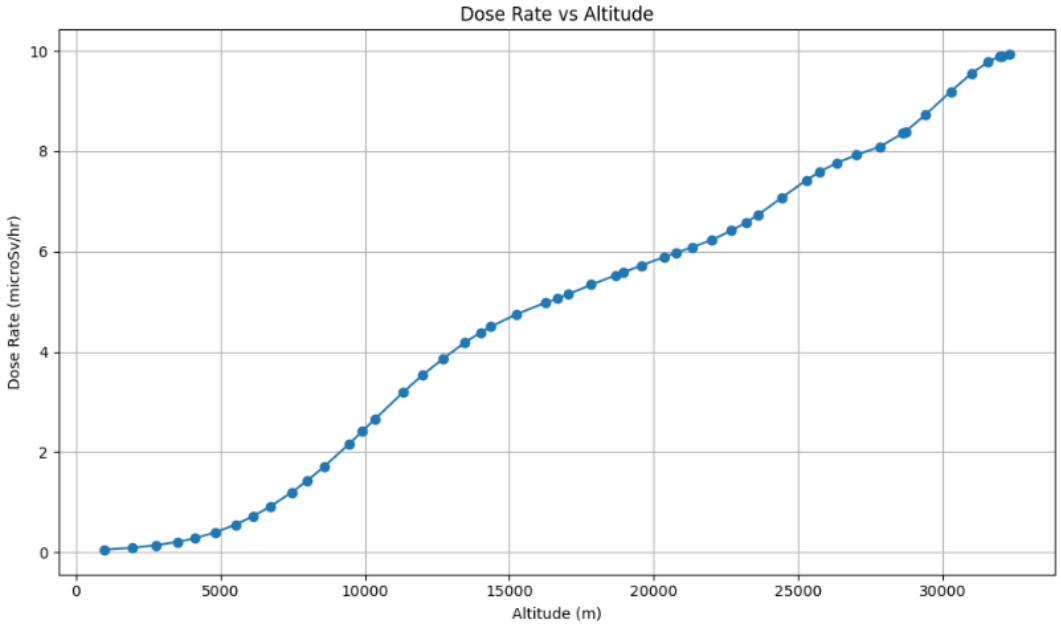


Figure 6:Average Dose Rate for 27/06/2024 at METU Calculated by CARI7-A.

One critical aspect of this research involves the choice of high-altitude balloons over sounding rockets. Sounding rockets, although useful for high-altitude measurements, face significant challenges due to their high acceleration, which complicates data collection during ascent. Furthermore, the cost of launching sounding rockets is considerably higher compared to high-altitude balloons, making balloons a more feasible option for repeated measurements and extended research campaigns [5] [9].

The comparison between high-altitude balloons and sounding rockets is illustrated by the altitude versus time graph from a high-altitude balloon flight and a typical sounding rocket flight. High-altitude balloons achieve their maximum altitude gradually, providing a stable platform for data collection, whereas sounding rockets reach their peak altitude rapidly, limiting the duration of usable data collection time at high altitudes.

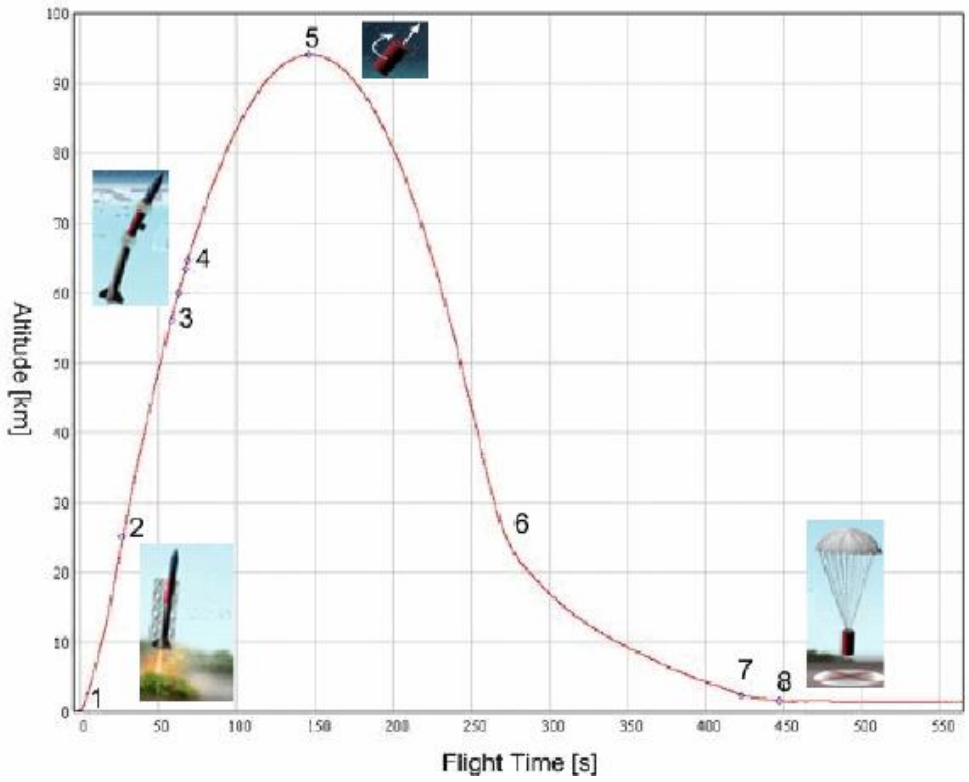


Figure 7:Altitude vs. Time Graph for Sounding Rockets [28]

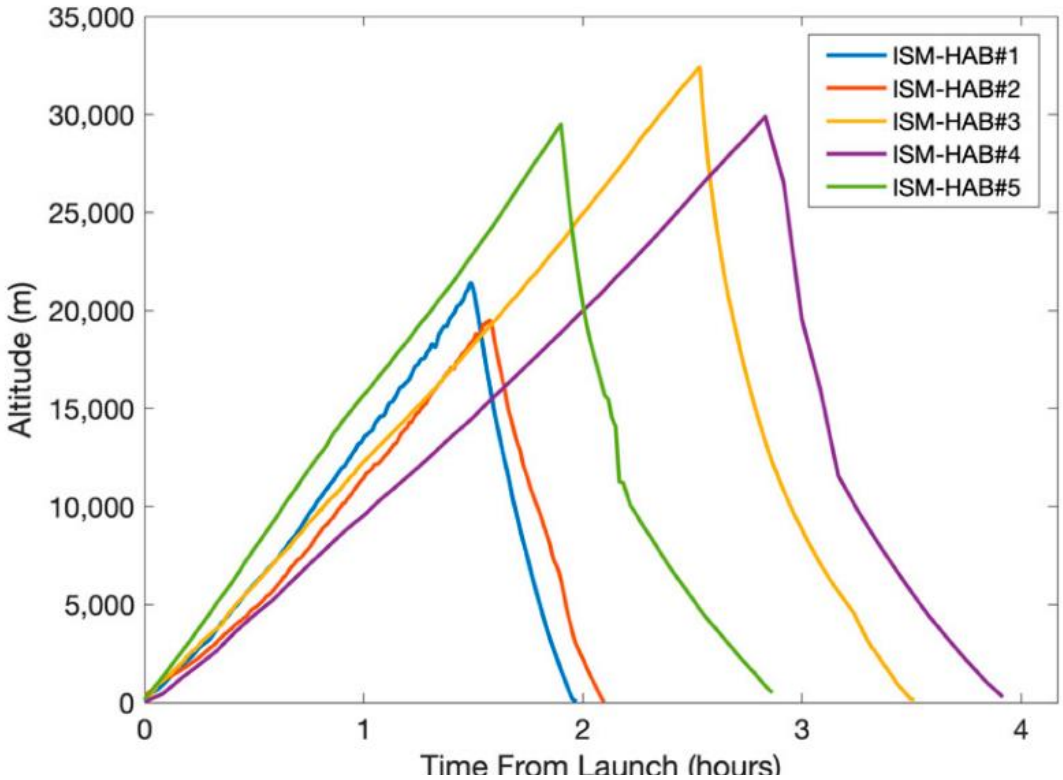


Figure 8:Altitude vs. Time Graph for Different High-Altitude Balloons [11]

The General Directorate of Meteorology makes observations at altitudes between 30,000 and 40,000 meters using high-altitude balloons. The aim of the research is to provide reusable measurement tools that work with these balloons. Several balloons are launched daily in Ankara by the General Directorate of Meteorology. The scope of the project is to create a product that can record mechanical and electronic activities, facilitate data transfer, and adapt to changes in flight trajectory during the mission, ensuring seamless communication throughout the flight.

The system consists of the following basic parts: a radiation sensor, electronic systems that provide landing, take-off, and data transfer, and a mechanical case. LoRa technology is essential for communication due to its long range and low power consumption. The electrical system includes an IMU and various sensors, including GPS and GPRS modules. A method called the Linear Kalman Filter is included to increase data precision, especially in the height estimation process.

In accordance with meteorology legislation, the updated system weight remained below 1.6 kg, while the useful load was limited to 1100 grams. The modification of the SB.01 radiation monitor prototype using only the Geiger circuit and the high voltage circuit and counter circuit of İvmeRad, developed at ODTÜ İVMER, is noteworthy and represents an important validation of the system's capabilities [9].

PROJECT SUMMARY

Battery Upgrade and Configuration:

After the interim report, the battery system was upgraded to 3200mAh Li-ion batteries arranged in a configuration of 3 parallel and 2 series, totaling 6 batteries. This configuration was chosen to ensure sufficient power supply for the entire system during high-altitude operation.

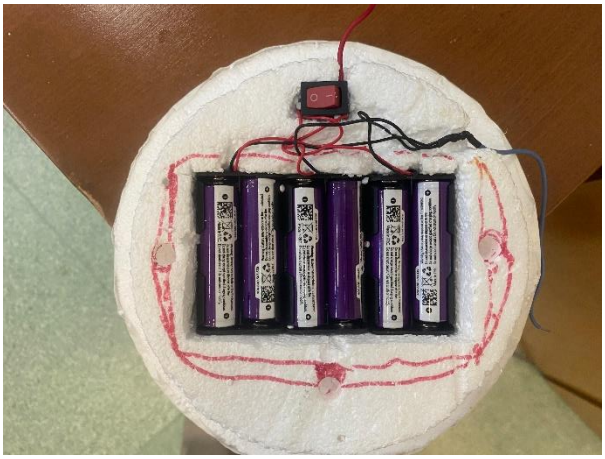


Figure 9: Li-ion Battery and Placement into Case

Relay and SD Card Heater:

A relay was added to enable and disable the SD card heater, which is controlled by a Dallas DS1820B temperature sensor attached to the SD card with tape. This relay replaces the previous buzzer relay, removing the buzzer from the system. The SD card heater is activated to maintain optimal temperature for the SD card during the flight.

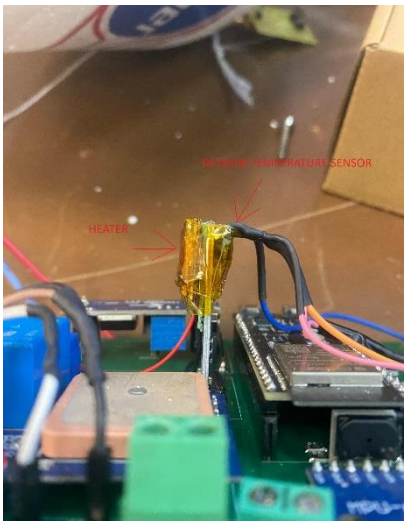


Figure 11: SD-Card Heater and Temperature Sensor



Figure 10: Relay Circuit of SD-Card Heater

Chassis Design:

A chassis was designed to improve cable management and stability of the electronic components. This chassis ensures that all components are securely fixed and organized, reducing the risk of disconnection or damage during flight.

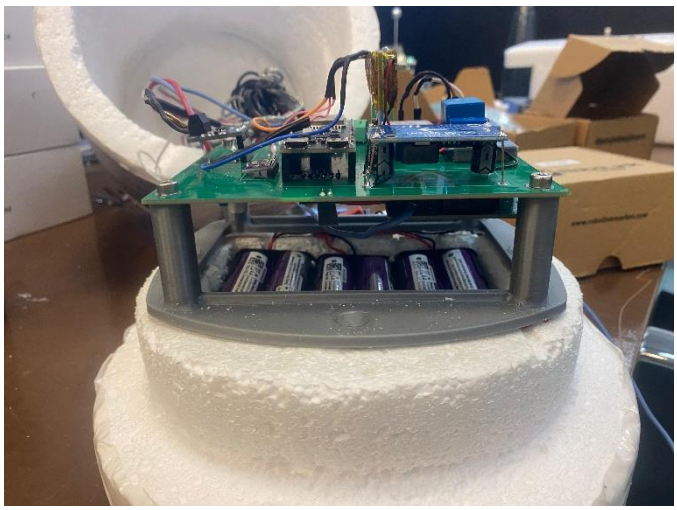


Figure 12: The Flight Card and Chassis Assembly

Geiger Tube and Antenna Placement:

The Geiger tube was placed outside the main case to ensure accurate radiation measurement. Additionally, antennas were extended outside the case using extension cables to enhance signal reception and transmission.

Temperature Measurement:

A K-type thermocouple was placed outside the case to measure the external temperature. This sensor provides data on the temperature conditions the system encounters during flight.



Figure 14: Payload Back Face and Geiger Tube and Antenna Placement



Figure 13: Payload Front Face and Temperature Measurement Setup

Ground Station Case:

A ground station case was designed and produced to house the ground station components securely. This case ensures protection and organization of the ground station equipment during operations.



Figure 16:Ground Station Circuit Board and the Case



Figure 15:Ground Station Enclosure

Antenna and Antenna Holder Design:

Antennas and their holders were designed and produced using 3D printing technology. These parts ensure stable and optimal placement of the antennas, improving communication reliability.



Figure 17: Antenna, Holder, and Ground Station Configuration

Balloon Configuration:

The balloon is filled with hydrogen gas and is designed to lift a payload up to 3300 kg. The system is connected as specified in the interim report:

- The balloon is attached to the parachute with a 4-meter string, providing stability during ascent and descent.
- A 26-meter rope is used for the flight configuration, consistent with the setup used by the General Directorate of Meteorology.
- The conical nose design of the case ensures aerodynamic stability.
- The entire system, including the mechanical casing and parachute, was configured to remain balanced during flight.



Figure 18:Balloon Launch Setup



Figure 19: Real-Time Data Transferring During Mission



Figure 20:Post-Flight Recovery of the Payload

RESULTS

The results of the high-altitude radiation measurement system are meticulously presented and analyzed, reflecting the systematic approach and detailed observations made during the test flights. The balloon, filled with hydrogen gas, embarked on its ascent and reached a maximum altitude of approximately 35,000 meters before descending back to the surface. The trajectory of the flight, depicted in the Altitude vs. Time graph, illustrates a gradual ascent over 5,500 seconds followed by a descent over the next 4,500 seconds. This stable trajectory facilitated comprehensive data collection during both phases of the flight.

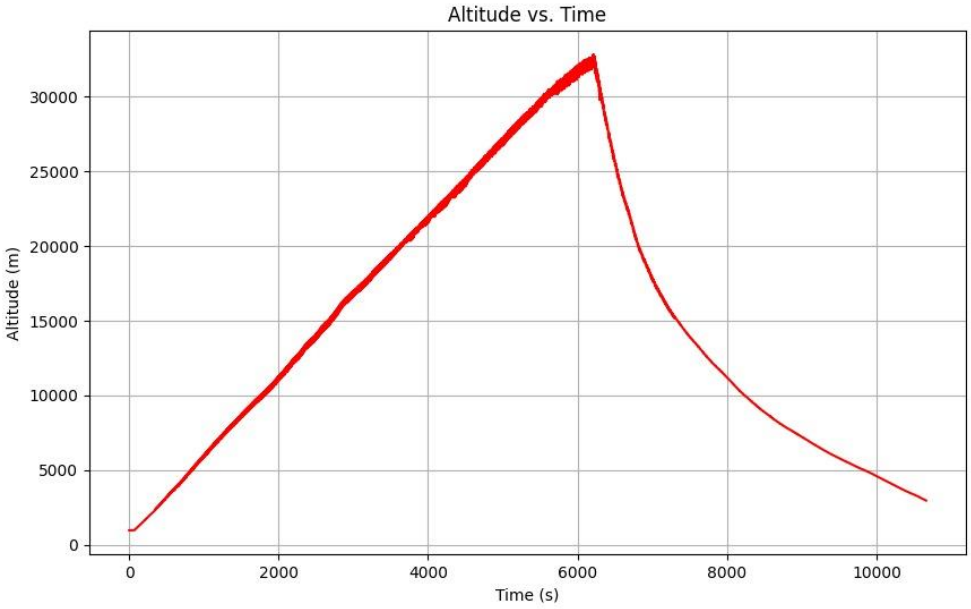


Figure 21:Altitude vs. Time Graph of the Balloon

Comparatively, the altitude-time graph from the Meteosis data shows that their balloon reportedly reached an altitude of 38,000 meters. However, it is important to note that Meteosis does not have descent data, which limits their ability to provide a complete profile of the flight. This absence of descent data could be due to various factors such as equipment limitations or data transmission issues during the descent phase. Despite reaching a higher altitude, the lack of descent information in the Meteosis data highlights a critical gap in their dataset, which our system has successfully addressed by providing a full trajectory profile including both ascent and descent phases.

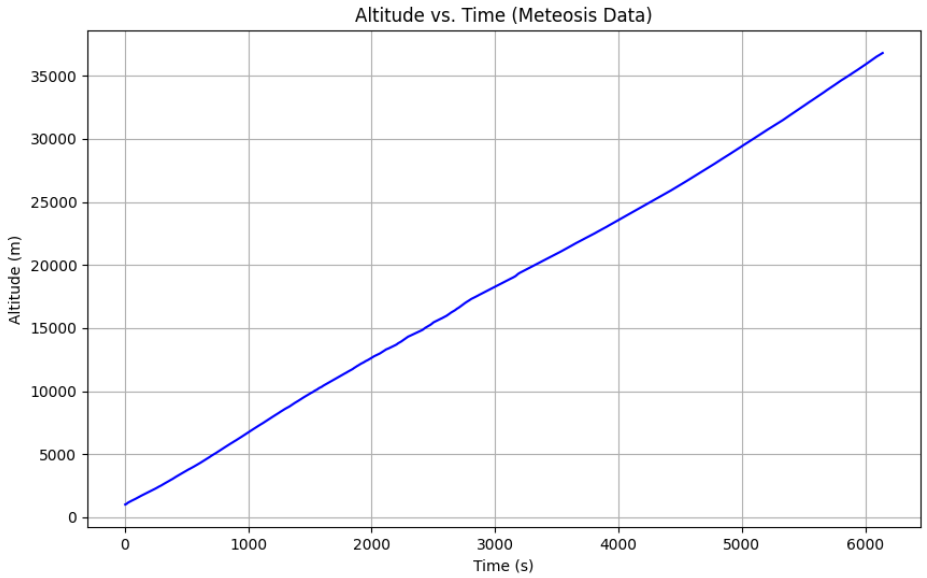


Figure 22:Altitude vs. Time Graph of the Balloon with Meteosis Altitude Data

Radiation measurements, represented by the mean Geiger counter counts recorded in 100-meter altitude intervals, revealed a clear increase in radiation levels with altitude. The counts peaked around 20,000 meters, corresponding to the Regener-Pfotzer maximum, characterized by the highest intensity of secondary cosmic radiation. Despite the subsequent malfunction of the Geiger counter, the data collected up to that point provides valuable insights into the radiation environment at high altitudes and underscores the system's potential for future research and applications in high-altitude radiation monitoring.

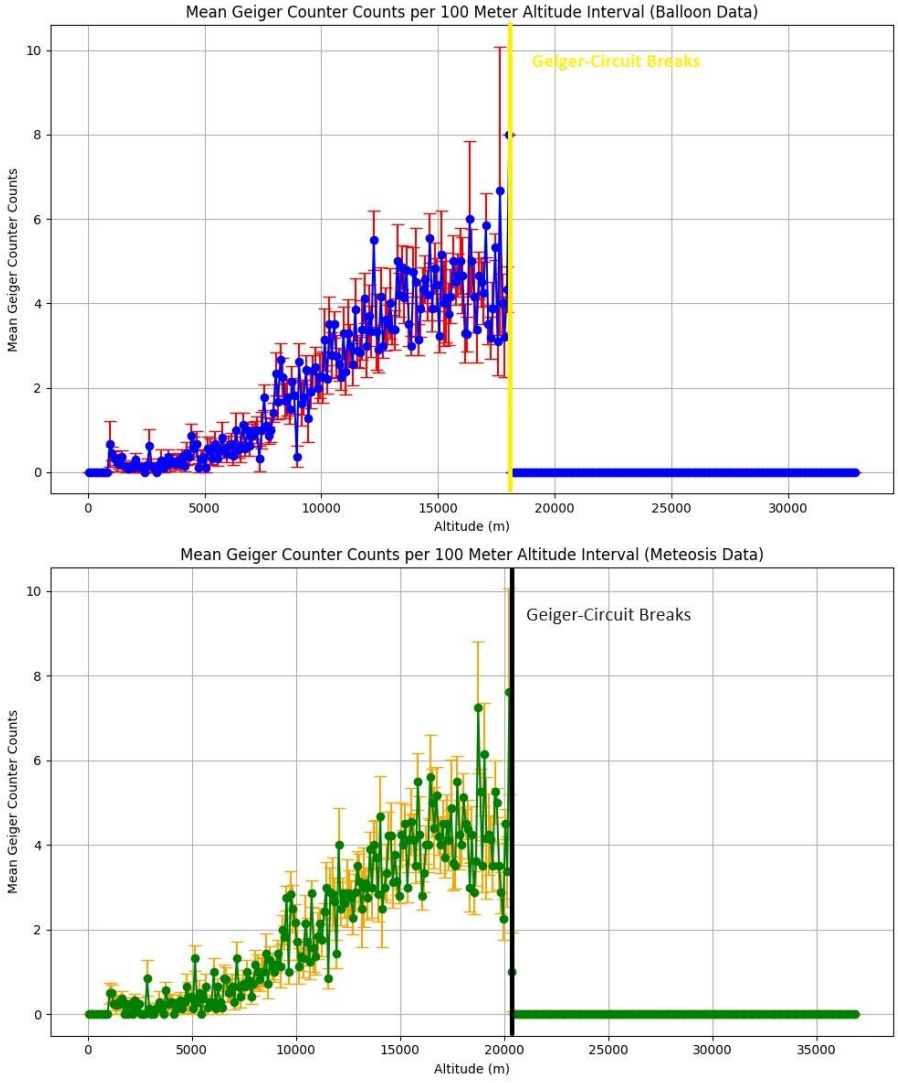


Figure 23:Mean Geiger Counter Counts per 100 Meter Altitude Interval Indicating the Regener-Pfotzer Maximum Recorded During the Balloon Flight Mission

The malfunction of the Geiger counter can be explained by the breakdown voltage phenomenon, which is depicted in the provided graph. The breakdown voltage decreases with decreasing pressure, and at high altitudes where the pressure is much lower, the distance between the legs of the transistor (1.4mm) was insufficient to prevent a spark. As shown in the graph, for air, the breakdown voltage can drop significantly at lower pressures, causing a spark and subsequent failure of the Geiger counter. By using this graph, we can say that the Geiger voltage breakdown happened, and the spark caused the transistor failure. After changing the transistor, everything functioned correctly.

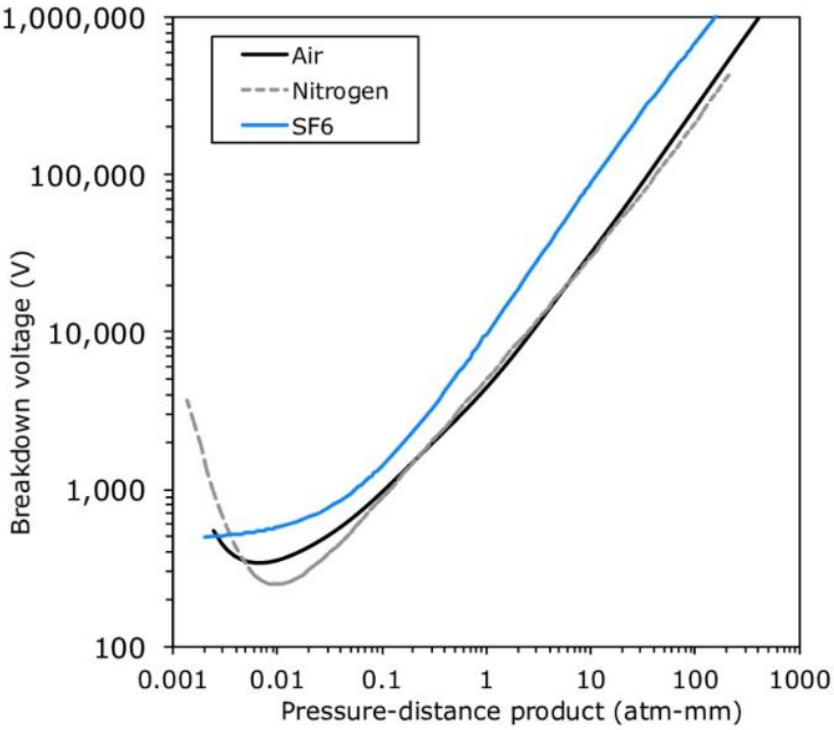


Figure 24: Pressure vs Breakdown Voltage of the air. (Paschen curve) [2]

This problem can be prevented by applying a conformal coating to the high-voltage components. Conformal coating is a protective chemical coating or polymer film that 'conforms' to the surface of an electronic component to protect it from moisture, dust, chemicals, and temperature extremes. By using conformal coating, the risk of high-voltage breakdown and sparking in low-pressure environments can be significantly reduced, ensuring the reliable operation of the Geiger counter at high altitudes.

The dose rate, measured in microsieverts per hour (μSv/hr), increases with altitude, reflecting the reduced atmospheric shielding at higher elevations. The data shows a significant increase in dose rate up to about 20,000 meters, after which it remains relatively stable. This trend is consistent with the expected increase in cosmic radiation exposure with altitude. However, the malfunction of the radiation sensor due to the high voltage spark at reduced atmospheric pressure limited the collection of dose rate data beyond this point. The graph underscores the importance of robust sensor design to withstand extreme high-altitude conditions for continuous and accurate data collection.

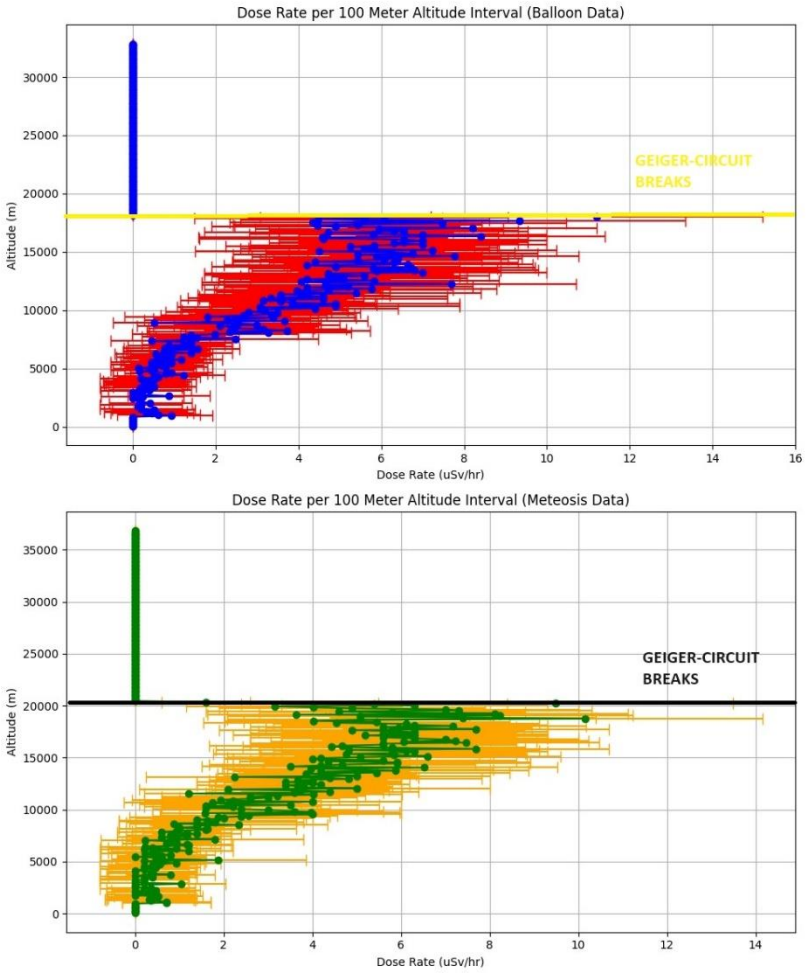


Figure 25Dose Rate per 100 Meter Altitude Interval with both Meteosis and Flight Card Altitude Data

Based on standard atmospheric radiation profiles and the data collected during this flight, the expected dose rates at various altitudes are as follows: at sea level, the dose rate is approximately 0.1 μSv/hr, increasing to around 10 μSv/hr at altitudes of 15-25 kilometers where the Regener-Pfotzer maximum occurs, and then decreasing again at higher altitudes. This correlation is vital for understanding the radiation environment at high altitudes and for developing radiation protection strategies for both high-altitude aviation and future space missions.

The observed data aligns well with the CARI-7A model (Figure 6), which is used for calculating cosmic radiation exposure. According to the CARI-7A data, dose rates increase significantly up to the Regener-Pfotzer maximum and then level off, similar to the trends observed in this experiment. This consistency validates the reliability of our data and the effectiveness of the CARI-7A model in predicting high-altitude radiation exposure.

Temperature measurements provided further insights into the atmospheric conditions encountered during the flight. The outside temperature decreased markedly with altitude during the ascent, starting from about 10°C at ground level and reaching approximately -50°C at 20,000 meters. Beyond 25,000 meters, the temperature stabilized around -50°C, aligning with expected atmospheric temperature gradients. The descent phase mirrored this trend with temperatures remaining low at higher altitudes and gradually increasing as the balloon approached the ground. These consistent temperature profiles underscore the reliability of the data collected.

Comparing our temperature data with that from Meteosis, we found close agreement in the outside temperature measurements. This comparison is depicted in the accompanying graphs, which show that both datasets follow a similar trend, thereby validating the accuracy of our temperature sensors and data collection methods.

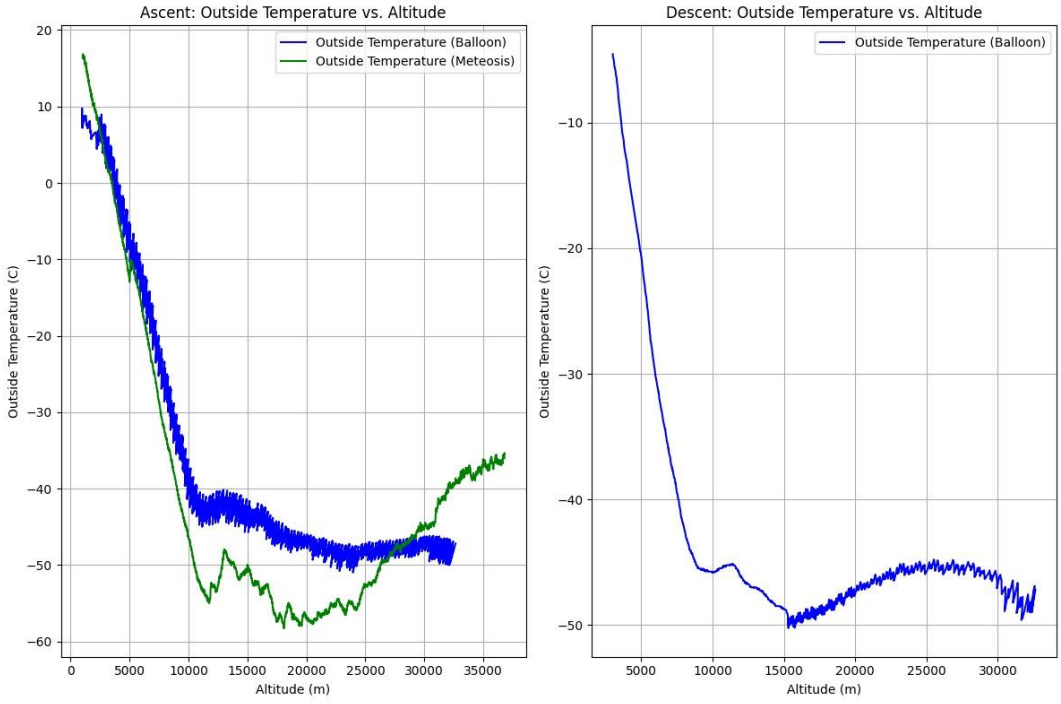


Figure 26:Temperature vs Altitude Recorded During the Balloon Flight

A critical aspect of the system's performance was its ability to maintain stable internal conditions for the payload electronics. The comparison of outside and inside temperatures during both ascent and descent phases demonstrated the effectiveness of the temperature control mechanisms. While the outside temperature fluctuated significantly, the inside temperature remained relatively stable until it dropped to -20°C. After this point, the heat isolation prevented the system from heating again. This stability is crucial for the integrity of the collected data and the overall success of the mission. Additionally, comparing our temperature data with that from Meteosis, we found close agreement in the outside temperature measurements, thereby validating the accuracy of our temperature sensors and data collection methods.

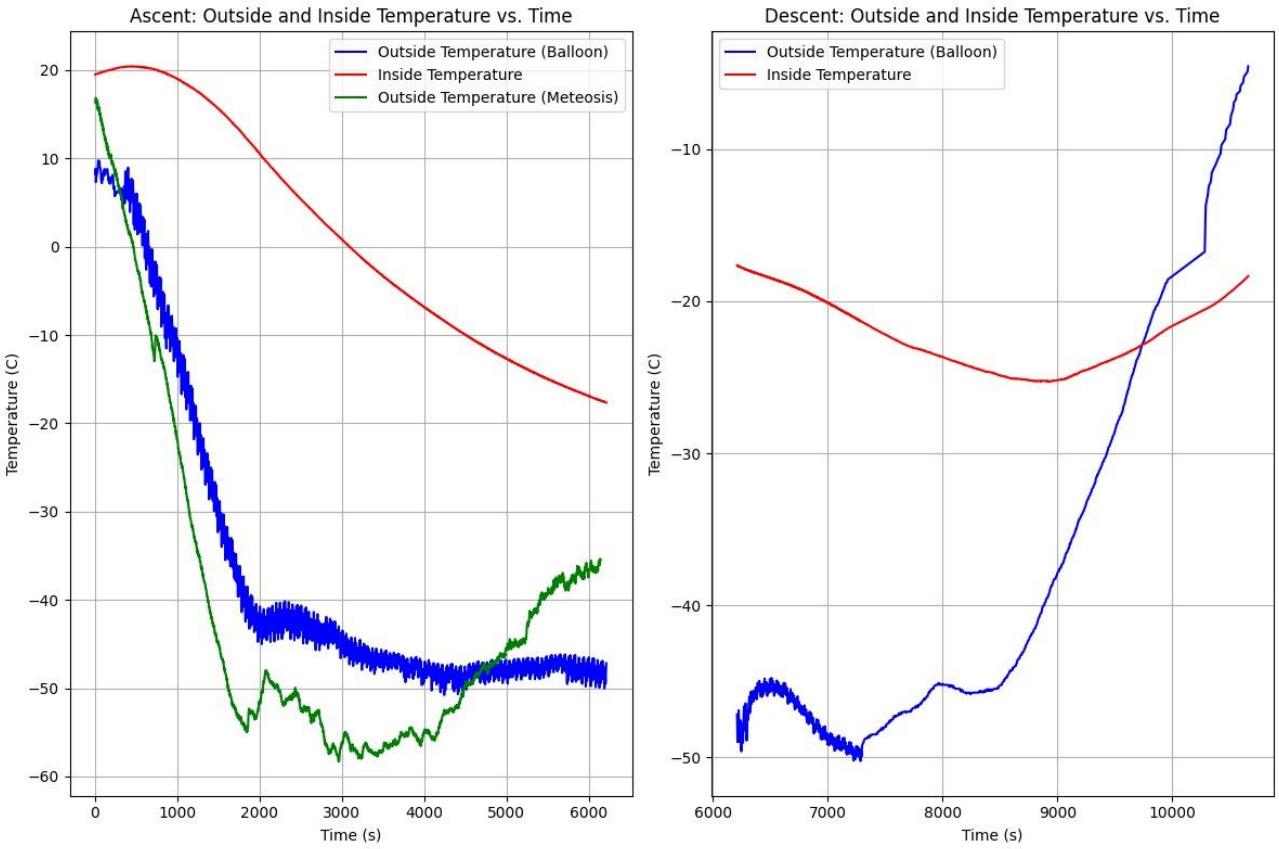


Figure 27:Comparison of Internal and External Temperature During the Balloon Flight

Humidity measurements were also taken throughout the ascent and descent, revealing a significant decrease in humidity levels with increasing altitude. During the ascent, humidity levels started at approximately 60% near the ground and dropped below 10% at altitudes above 15,000 meters. The descent phase exhibited a similar trend, albeit with slightly higher humidity levels at corresponding altitudes, likely due to varying atmospheric conditions during different phases of the flight. These findings are consistent with standard atmospheric humidity profiles, validating the accuracy of the sensors used.

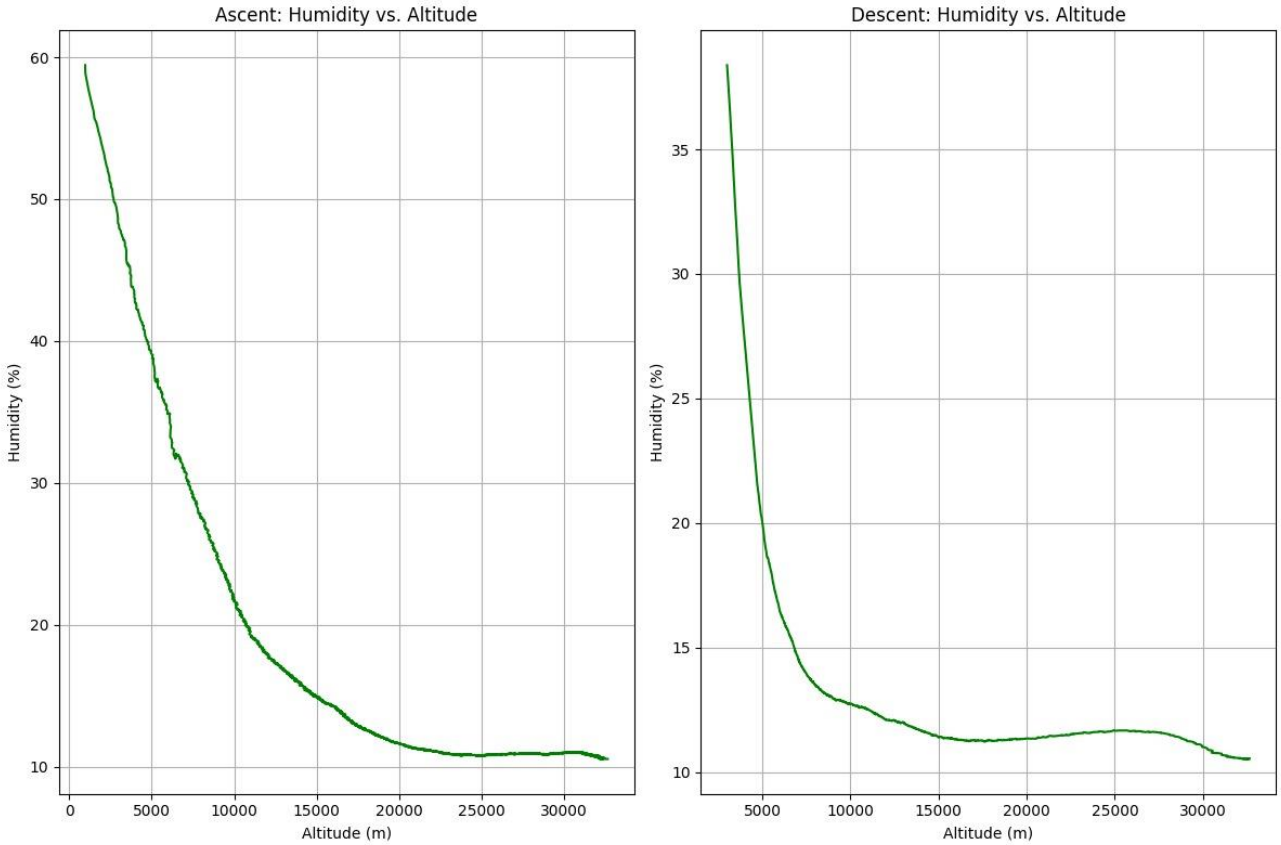


Figure 28:Humidity vs. Altitude Recorded During the Balloon Flight

In summary, the high-altitude radiation measurement system exhibited robust performance across various parameters. The balloon achieved a maximum altitude of approximately 30,000 meters, providing a stable platform for comprehensive data collection. The observed humidity and temperature trends were consistent with standard atmospheric profiles, confirming the accuracy of the sensors and data collection methodologies. The internal temperature control system effectively maintained stable conditions for the payload electronics, ensuring the integrity of the collected data. The increase in particle counts with altitude, peaking around 20,000 meters, validates the system's capability in measuring high-altitude radiation. These results confirm the efficacy of the reusable

high-altitude radiation measurement system, showcasing its potential for providing valuable insights into atmospheric and cosmic radiation profiles and paving the way for future research and applications in this field.

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